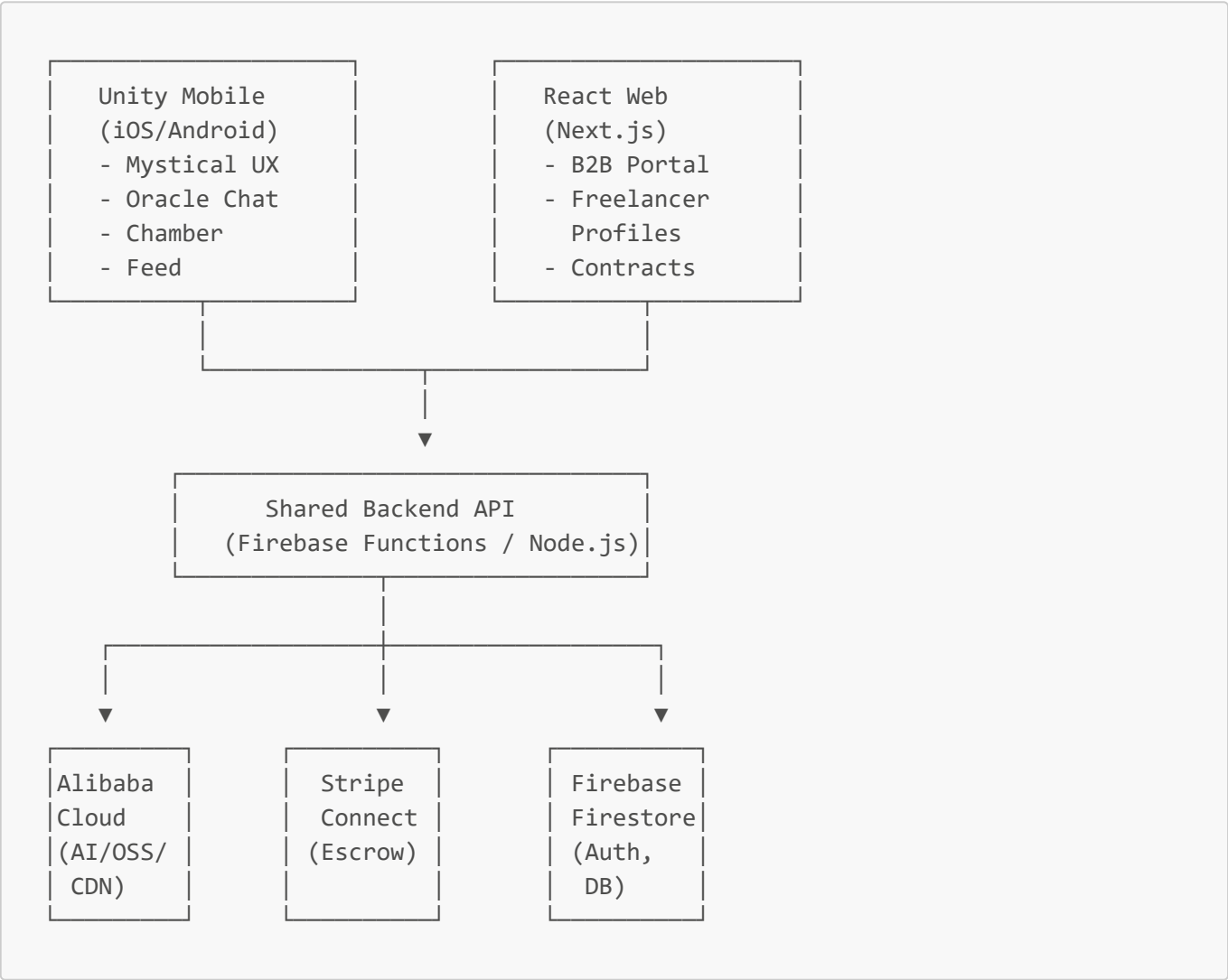


TIMEAURA TECHNICAL ARCHITECTURE

Executive Summary

TimeAura is a hybrid platform consisting of a Unity mobile app (iOS/Android) for the mystical Nexus experience, and a React/Next.js web portal for professional B2B interactions, freelancer profiles, and contract management. Built to target freelancers, creators, designers, and developers who trade expertise measured in **Horas** (time credits). Monetization combines a premium **Enlightened** subscription (Unity IAP) and a small commission on transactions (Stripe Connect). The unique value proposition lies in an AI-driven Oracle that acts as a community participant, powered by a three-layer economy (Fiat ↔ Horas ↔ Premium).

System Architecture



Platform Architecture

TimeAura operates as a hybrid platform with two client interfaces sharing a unified backend:

Mobile App (Unity)

- **Purpose:** Mystical "Nexus" experience with AI Oracle

- **Target users:** Active freelancers, daily interactions
- **Key features:**
 - Oracle Eye (floating AI widget)
 - Sanctuary Panel (full AI chat)
 - Convergence Feed (service requests)
 - Chamber (active pacts workspace)
 - Voice intent parsing
- **Technology:** Unity 6.3 LTS (6000.3.10f1), UI Toolkit, VContainer

Web Portal (React/Next.js)

- **Purpose:** Professional B2B interactions and contract management
- **Target users:** Business clients, freelancers managing profiles
- **Key features:**
 - Freelancer profiles with portfolios
 - Contract creation and management
 - Escrow dashboard
 - Invoice generation
 - Admin panel for TimeAura team
 - SEO-optimized public pages (for Google indexing)
- **Technology:** Next.js 14, React 18, TypeScript, Tailwind CSS

Shared Backend

- **API Layer:** Firebase Cloud Functions (or Node.js Express)
- **Database:** Firebase Firestore (NoSQL) + ApsaraDB (relational for contracts)
- **Authentication:** Firebase Auth (unified across mobile + web)
- **Storage:** Alibaba Cloud OSS (user-generated content)
- **AI:** Alibaba Cloud DashScope (Qwen) — accessible from both platforms

AI Integration Layer

Current integration

- **IOracleService** – unified interface for all LLM interactions.
- **Providers:**
 - **QwenOracleProvider** – primary provider using Alibaba DashScope (optimized for Slavic languages).
 - **GeminiOracleProvider** – fallback provider (Google Gemini).
- **OracleServiceProvider** – router that selects provider based on language availability and health.
- **Use-cases:**
 - Voice intent parsing (Voice → Text → Intent).
 - Interactive chat in the Sanctuary panel.
 - Content generation (service descriptions, bios).
 - Conflict arbitration in the Chronos Court.

Architectural benefits

- **Modular** – new providers (e.g., Grok, OpenAI) can be added without UI changes.
- **Automatic fallback** – seamless switch when the primary provider is unavailable.
- **Localization aware** – tone adapts per-user language settings.

Planned Generative Models

1. **Image Generation** (Tongyi Wanxiang / Stable Diffusion)
 - Unique user avatars generated from Aura tags.
 - Service icons for categories (e.g., plumbing, code, design).
 - Illustrated feed posts with a mystical/RPG visual style.
2. **Video Generation**
 - Auto-created tutorials for onboarding newcomers.
 - Short promotional clips for marketing campaigns.
 - UI animation assets (spell effects, transitions).
3. **Audio Generation**
 - Mystical sound effects for UI interactions.
 - Ambient background music for different panels.
 - Text-to-Speech for Oracle responses with a “seer” voice.
4. **Multimodal AI**
 - **Voice + Vision:** user shows a problem image while speaking → AI analyses both streams.
 - **Text + Image:** generate service description from an uploaded picture.

Infrastructure & Scaling

Current stack

- **Client:** Unity Mobile (iOS/Android) with UI Toolkit.
- **Config:** Google Sheets Remote Config (temporary).
- **Database:** Firebase Firestore (profiles, posts, deals).
- **Payments:** Stripe Connect (fiat) + Unity IAP (subscriptions).

Planned Alibaba Cloud expansion

- **AI Models:** DashScope APIs (Qwen-Max/Plus for text, Tongyi Wanxiang for images).
- **Storage:** OSS (Object Storage Service) for generated assets (avatars, icons, videos).
- **CDN:** Alibaba Cloud CDN for global, low-latency delivery.
- **Compute:** Function Compute for server-side logic (escrow, arbitration, validation).
- **Database:** ApsaraDB (RDS) as a fallback when Firestore scaling limits are reached.

Scaling roadmap

Phase	Mobile Users	Web Users	Tokens/Day	Storage
Soft Launch	1k	200	50k	10 GB
Full Launch	10k	2k	500k	100 GB
Creator Platform	100k	20k	5M	1 TB

Technology Stack

Client — Mobile

- Unity 6.3 LTS (6000.3.10f1)
- UI Toolkit (UXML/USS)
- C# (VContainer for DI)
- Unity IAP (subscriptions)

Client — Web (NEW)

- Next.js 14 (React framework)
- TypeScript
- Tailwind CSS (styling)
- React Query (data fetching)
- Stripe.js (payments)

Shared Backend

- Firebase (Firestore, Auth, Cloud Functions)
- Stripe Connect (payments)
- Google Sheets API (configuration)
- Node.js (API layer for web)

AI/ML

- Alibaba Cloud DashScope (Qwen-Max, Qwen-Plus — text)
- Google Gemini (fallback LLM)
- Planned: Tongyi Wanxiang (image), Video/Audio generation models

Infrastructure (Alibaba Cloud)

- OSS (object storage)
- CDN (content delivery)
- Function Compute (serverless logic)
- ApsaraDB RDS (relational database)
- API Gateway (unified API for mobile + web)

Prepared for Alibaba Cloud partnership discussion.